

YANSHU ZHANG

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RESEARCH INTERESTS

I build interactive virtual worlds from images, spanning neural rendering, point-based 3D representations, feed-forward novel view synthesis, 3D reconstruction, editing and animation, and generative models.

EDUCATION

Simon Fraser University 09/2021 - Expected 04/2027

Ph.D. in Computer Science (CGPA: 4.26/4.3), supervised by Prof. Ke Li

University of Science and Technology of China 09/2017 - 07/2021

B.S. in Computer Science

RESEARCH EXPERIENCE

APEX Lab, Simon Fraser University — Ph.D. Researcher 09/2021 - Present

- Led PAPR (NeurIPS 2023 Spotlight), a point-based neural renderer using ray-to-point proximity attention that deforms without the holes seen in 3D Gaussian Splatting.
- Extended PAPR to geometry/texture editing (PointGT, P-CORE; ECCV 2026), close-up rendering (3DV 2026), 4D interpolation (CVPR 2024 Highlight), and rig-based animation (NeurIPS 2026).
- Built PointWeave, a feed-forward renderer with explicit 3D points and ray-to-point attention: +1.6 dB PSNR over 3DGS-based methods with fewer primitives; up to 4× faster than geometry-free models.
- Contributed to generative-model evaluation (a Chamfer-distance metric; NeurIPS 2026) and uncertainty-aware world models for RL (WIMLE; ICLR 2026).

National University of Singapore — Research Exchange Programme (non-degree) 01/2021 - 05/2021

PUBLICATIONS

PointWeave: Feed-Forward View Synthesis with Explicit Point Interface

Yanshu Zhang, Shichong Peng, Chirag Vashist, Ke Li Under Submission

RigPAPR: Rig-Based Animation of Static Neural Point Clouds from a Single-View Video

Shichong Peng, **Yanshu Zhang**, Ke Li NeurIPS 2026

The Fault in Our Metrics: Revisiting Generative Model Evaluation with Chamfer Distance

Chirag Vashist, Vaibhav Santhosh, Tristan Engst, **Yanshu Zhang**, Ke Li NeurIPS 2026 E&D

PointGT: Simultaneous Geometry and Texture Editing for Point-Based Representations

Yanshu Zhang, George Shramko, Pratul P. Srinivasan, Ke Li ECCV 2026

P-CORE: Self-Supervised Surface Consistency for Point-Based Neural Editing

Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li ECCV 2026

Tackling Misattribution in 3D Intrinsic Decomposition via Proximity Attention Point Rendering

Alireza Moazeni, Shichong Peng, **Yanshu Zhang**, Chirag Vashist, Ke Li ECCV 2026

PAPR Up-close: Close-up Neural Point Rendering without Holes

Yanshu Zhang, Chirag Vashist, Shichong Peng, Ke Li 3DV 2026

WIMLE: Uncertainty-Aware World Models with IMLE

Mehran Aghabozorgi, Alireza Moazeni, **Yanshu Zhang**, Ke Li ICLR 2026

PROSE: Point Rendering of Sparse-Controlled Edits to Static Scenes

Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li *ECCV 2024 Workshop*

PAPR in Motion: Seamless Point-level 3D Scene Interpolation

Shichong Peng, **Yanshu Zhang**, Ke Li

CVPR 2024 (Highlight)

PAPR: Proximity Attention Point Rendering

Yanshu Zhang*, Shichong Peng*, Alireza Moazeni, Ke Li

NeurIPS 2023 (Spotlight)

PATENTS

Method and System for Accelerated Operation of Layers Used in a Machine Learning Model and Differentiable Point Rendering Using Proximity Attention

Yanshu Zhang, Yuzhen Mao, Shichong Peng, Alireza Moazeni, Ke Li

CA Patent Application: 3292617 (2024)

TECHNICAL SKILLS

Programming: Python, C++, CUDA (custom kernels)

Frameworks & Tools: PyTorch, multi-GPU distributed training (DDP), SLURM clusters

Research Areas: Neural rendering, NeRF / 3D Gaussian Splatting, point-based rendering, novel view synthesis, 3D editing and animation, generative models

AWARDS

SFU Ph.D. Research Scholarship (*2022, 2024*)

SFU Graduate Fellowship (*2023, 2025*)

Silver Award for Outstanding Students of USTC (*2019*)

Bronze Award for Outstanding Students of USTC (*2018*)

TEACHING EXPERIENCE

Generative Models, Teaching Assistant, SFU (*2025*)

Machine Learning, Teaching Assistant, SFU (*2022*)

ACADEMIC SERVICE

Served as reviewer for CVPR, ECCV, ICCV, 3DV, SIGGRAPH, SIGGRAPH Asia, ICML, ICLR, NeurIPS